

Maheer Ihtejaz Moin

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SUMMARY

Bachelor of Advanced Computing (Honours) student at the University of Sydney majoring in Computational Data Science and Finance, with exceptional programming and design skills across Python, C++, and Java. Hands-on experience building high-performance, resilient platforms, large-scale data research pipelines, and web frameworks, applying machine learning, distributed computing, probability and statistics, and natural language processing. Strong analytical and problem-solving skills with a passion for solving challenging problems in collaborative, highly technical teams. Seeking a software engineering or quantitative internship to design systems architectures and turn ideas into production-grade technology across software, data, and quantitative finance.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, R

Software Engineering: systems architecture, high-performance & low-latency systems, distributed computing, concurrency, data structures & algorithms, object-oriented programming, Linux, Git/GitHub, Docker, Kubernetes, CI/CD (GitHub Actions), Pytest, Agile / short development cycles

Web & Platforms: React, Node.js, Tailwind CSS, Supabase, Streamlit, REST APIs, full-stack development

Machine Learning & Data: machine learning, natural language processing, large language model (GPT-4) API integration, XGBoost, NumPy, Pandas, SciPy, large data research platforms, statistical modelling

Quantitative & Finance: probability & statistics, time-series forecasting (ARMA-GARCH, HAR-RV), Monte Carlo simulation, hypothesis testing, mathematical modelling, quantitative research, options pricing, portfolio optimisation, market microstructure, risk management

EDUCATION

The University of Sydney | Sydney, NSW | 2024 - 2028

Bachelor of Advanced Computing (Honours), Majors: Computational Data Science & Finance

- Dalyell Scholar | Vice Chancellor's Scholarship

WORK EXPERIENCE

2A Marketing | Sydney, NSW | Mar 2023 - Dec 2023

- Built and deployed a full-stack website for company information, communications, and online sales, showcasing my skills as a full-stack engineer, featuring a splash screen and Framer Motion animations for a polished, responsive user experience, hosted on the company domain.
- Architected the backend on Supabase for data storage, authentication, and live content management.
- Built a shipment-tracking portal emulating a version-control-style environment to manage shipping documents, with live shipment status updates.

PROJECTS

Future Volatility Forecasting | *Optiver Industry Project* | Python, XGBoost

- Engineered features from large-scale order-book data across 10 technology equities (realised volatility, log returns, bid-ask spread, WAP) and fitted HAR-RV and ARMA-GARCH models with a trained XGBoost regressor to forecast future stock volatility.
- Built a trading-engine application that uses the fitted and trained models to predict future stock volatility on the provided dataset and drive an options-pricing engine.

Stock Portfolio Optimisation | *Markowitz Mean-Variance Model* | Python, SciPy

- Implemented Modern Portfolio Theory with quadratic optimisation (SciPy) to derive the efficient frontier and maximum-Sharpe-ratio / minimum-variance portfolios, validated with Monte Carlo simulation.
- Built an interactive application around the optimiser, showcasing end-to-end software engineering from quantitative model to a usable, deployable tool.

End-to-End Automated Data Analysis Pipeline | Python, Streamlit, GPT-4 API

- Built a fully automated pipeline from CSV upload to insight generation: cleaning data, handling outliers and missing values, and running inferential tests (t-tests, Levene's test) with no manual steps.
- Integrated the GPT-4 (natural language processing) API to translate statistical results into human-readable summaries, delivered through an interactive Streamlit web app.

Efficient Email Filtering System | *Bloom Filter* | C++

- Engineered a space-efficient 7-hash Bloom filter (15,000-bit array, custom polynomial rolling hashes) for low-memory duplicate detection with zero false negatives; added Base64 serialisation, CLI parsing, and file/live ingestion for real-time ADD/CHECK/SHOW.